

Miscellaneous Theorems and Ideas

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This is just a collection of some cool facts and theorems I have found, along with their proofs.

Theorem 1.

For an odd positive integer n , and positive integers a, b with $a^2 \leq b < (a+1)^2$ we have the following,

$$\left\lfloor (a + \sqrt{b})^n \right\rfloor = (a + \sqrt{b})^n + (a - \sqrt{b})^n.$$

Proof of Theorem 1.

Firstly, due to Binomial expansion,

$$\begin{aligned} (a + \sqrt{b})^n + (a - \sqrt{b})^n &= \sum_{r=0}^n \binom{n}{r} (\sqrt{b})^r a^{n-r} + \sum_{r=0}^n \binom{n}{r} (-\sqrt{b})^r a^{n-r} \\ &= \sum_{r=0}^n \binom{n}{r} (\sqrt{b})^r [1 + (-1)^r] a^{n-r} \\ &= 2 \sum_{\substack{r=0 \\ r \text{ even}}}^n \binom{n}{r} (\sqrt{b})^r a^{n-r} \end{aligned}$$

which is clearly an integer. Now it suffices to show that

$$(a + \sqrt{b})^n - 1 < (a + \sqrt{b})^n + (a - \sqrt{b})^n \leq (a + \sqrt{b})^n$$

That is

$$-1 < (a - \sqrt{b})^n \leq 0$$

but we know that $a \leq \sqrt{b} < a+1$ and so $a - \sqrt{b} \leq 0$ and $-1 < a - \sqrt{b}$ and raising both inequalities to the power of n gives the desired result. ■

Theorem 2.

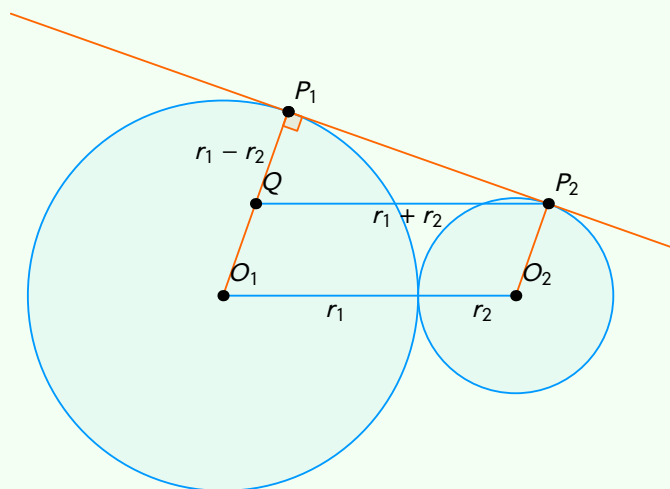
Let C_1 and C_2 be externally tangent circles with centres O_1, O_2 and radii r_1, r_2 , ($r_1 \geq r_2$) respectively. Let P_1, P_2 be points on C_1, C_2 respectively such that P_1P_2 is a common tangent to both circles.

(a) Show that $P_1P_2 = 2\sqrt{r_1r_2}$.

(b) Find the area of the region bounded by the two circles and their common tangent.

Solution

(a) The idea is to draw in the line l through P_2 that is parallel to O_1O_2 . Let $Q = l \cap O_1P_1$.



We notice that since O_1P_1 and O_2P_2 are radii they meet the common tangent at right angles, and in particular $O_1P_1 \parallel O_2P_2$. Now it is clear to see that $QP_2 = r_1 + r_2$ and $QP_1 = r_1 - r_2$. Now owing to Pythagoras,

$$P_1P_2 = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = 2\sqrt{r_1r_2}. \quad \square$$

(b) Notice that due to the parallel conditions $O_1P_1P_2O_2$ is a trapezium, with area

$$\frac{r_1 + r_2}{2} (2\sqrt{r_1r_2}).$$

Now we just need to subtract off the two sectors. Let $\angle P_1O_1O_2 = \theta$. Notice that $\angle O_1O_2P_2 = 180 - \theta$ due to corresponding angles. Taking θ to be measured in radians gives the area of the two sectors to be

$$\frac{1}{2}r_1^2\theta + \frac{1}{2}r_2^2(\pi - \theta) = \frac{1}{2}\theta(r_1^2 - r_2^2) + \frac{1}{2}\pi r_2^2.$$

So the final area is

$$(r_1 + r_2)\sqrt{r_1r_2} - \frac{1}{2}\theta(r_1^2 - r_2^2) + \frac{1}{2}\pi r_2^2. \quad \blacksquare$$

Comments: I first came across the fact (a) from a BPRP (Black pen red pen) video, and then I actually used this fact to solve a problem in a Year 12/13 geometry topic test. Part (b) actually came up as a question with special case $r_1 = 3, r_2 = 1$ when I was sitting the 2026 Jan TMUA, I was unable to solve it during contest.

Theorem 3.

Consider some 3×3 matrix A of the form

$$A = \begin{bmatrix} k & 0 & 0 \\ x & k & 0 \\ y & z & k \end{bmatrix}.$$

Let M be another 3×3 matrix defined like so, $A = kI + M$. Then

$$A^n = k^n I + k^{n-1} nM + k^{n-2} \frac{n(n-1)}{2} M^2.$$

Solution

Claim 1: M is nilpotent with nilpotency index ≤ 3 . In other words there is a positive integer $k \leq 3$ such that $M^k = 0$.

Proof: We use the Caley-Hamilton theorem. Look at the characteristic equation of M ,

$$\begin{vmatrix} -\lambda & 0 & 0 \\ x & -\lambda & 0 \\ y & z & -\lambda \end{vmatrix} = -\lambda^3 = 0.$$

Now since M satisfies it's own characteristic equation (by Caley-Hamilton) we have $M^3 = 0$. $\square$

Now we can apply Binomial theorem like so,

$$\begin{aligned} A^n &= (kI + M)^n \\ &= \sum_{r=0}^n \binom{n}{r} (M)^r (kI)^{n-r} \\ &= \binom{n}{0} k^n I + \binom{n}{1} k^{n-1} M + \binom{n}{2} k^{n-2} M^2 \\ &= k^n I + n k^{n-1} M + \frac{n(n-1)}{2} k^{n-2} M^2 \end{aligned}$$

where the expansion stops at $r = 2$ because any further terms will have a factor of $M^3 = 0$. $\blacksquare$

Theorem 4.

Take $2^n + 1$ lattice points in n -dimensional space. Then there are two points P, Q out of this group such that the segment with endpoints P, Q passes through another point with integer coordinates.

Solution

Consider a point from this group

$$P = (p_1, p_2, \dots, p_n).$$

There are 2^n ways to choose each of the parities of each of the $p_i, (i = 1, 2, \dots, n)$. Since we have $2^n + 1$ points, by Pigeonhole principle there must be some point Q also a part of this group

$$Q = (q_1, q_2, \dots, q_n)$$

that satisfies

$$p_i \equiv q_i \pmod{2}, \quad (i = 1, 2, \dots, n).$$

Now define

$$\vec{\Delta} = \vec{OQ} - \vec{OP} = \langle x_1, x_2, \dots, x_n \rangle$$

where $x_i = q_i - p_i$. Note that x_i is even because p_i and q_i have the same parity. Since every component is even

$$\frac{1}{2}\vec{\Delta} \in \mathbb{Z}^n$$

and so the point at

$$\vec{OP} + \frac{1}{2}\vec{\Delta} = \left(p_1 + \frac{1}{2}x_1, p_2 + \frac{1}{2}x_2, \dots \right)$$

is a lattice point and lies on the segment PQ between P and Q . ■

Theorem 5 — (Strengthened Theorem 1).

Let $a, b \in \mathbb{N}$, b not a perfect square. Define $x = a + \sqrt{b}$ and $y = a - \sqrt{b}$. Let $n \in \mathbb{N}$.

- If $0 < y < 1$, then $x^n + y^n = \lceil x^n \rceil$.
- If $-1 < y < 0$ and n is odd, then $x^n + y^n = \lfloor x^n \rfloor$.

Solution

We present a strengthened version of theorem 1, along with motivation for why these statements are true. Define $S_n = x^n + y^n$. First we are going to show that $S_n \in \mathbb{Z}$. Then since $|y| < 1$ we know that y^n is very small, enough to restrict S_n to an interval of size < 1 . Finally we are able to conclude the results of theorem 5.

Note that x, y are the roots of

$$x^2 - 2ax + (a^2 - b) = 0.$$

If one has studied how to solve second order linear recurrence relations then you know that it is the case that

$$S_{n+1} = 2aS_n - (a^2 - b)S_{n-1}$$

with the initial conditions $S_0 = 2, S_1 = 2a$ both of which are integers. Since $a, b \in \mathbb{N}$ and the integers are closed under $+, -, \times$ we deduce that $S_n \in \mathbb{Z}$.

Now observe that if $0 < y < 1$ then $0 < y^n < 1$. Thus S_n satisfies the inequality

$$x^n < S_n < x^n + 1.$$

Now since S_n is an integer in the interval $(x^n, x^n + 1)$ it is the smallest integer $> x^n$, that is $S_n = \lceil x^n \rceil$.

Similarly if $-1 < y < 0$ then $-1 < y^n < 0$ since n is odd. So S_n satisfies

$$x^n - 1 < S_n < x^n.$$

Again since S_n is an integer in the interval $(x^n - 1, x^n)$ it is the largest integer $< x^n$, that is $S_n = \lfloor x^n \rfloor$. ■